CELL STRUCTURE

The cell is the basic unit of life. Every living thing is made of cells.

The nucleus controls the cell and contains DNA.

Mitochondria are the powerhouse of the cell; they release energy from glucose.

The cell membrane controls what enters and leaves the cell.

Photosynthesis

Green plants make food using sunlight, carbon dioxide and water.

Chlorophyll in the chloroplasts absorbs light energy.

The products are glucose and oxygen.

$6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$

Respiration

Respiration releases energy from glucose in the mitochondria.

Aerobic respiration needs oxygen and makes carbon dioxide and water.

Anaerobic respiration in muscles makes lactic acid.